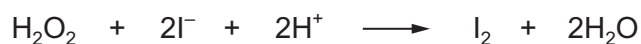


12. (a) Iodide ions can be oxidised to iodine by reaction with acidified hydrogen peroxide.



The rate of reaction can be followed in a clock reaction by the appearance of a blue-black colour.

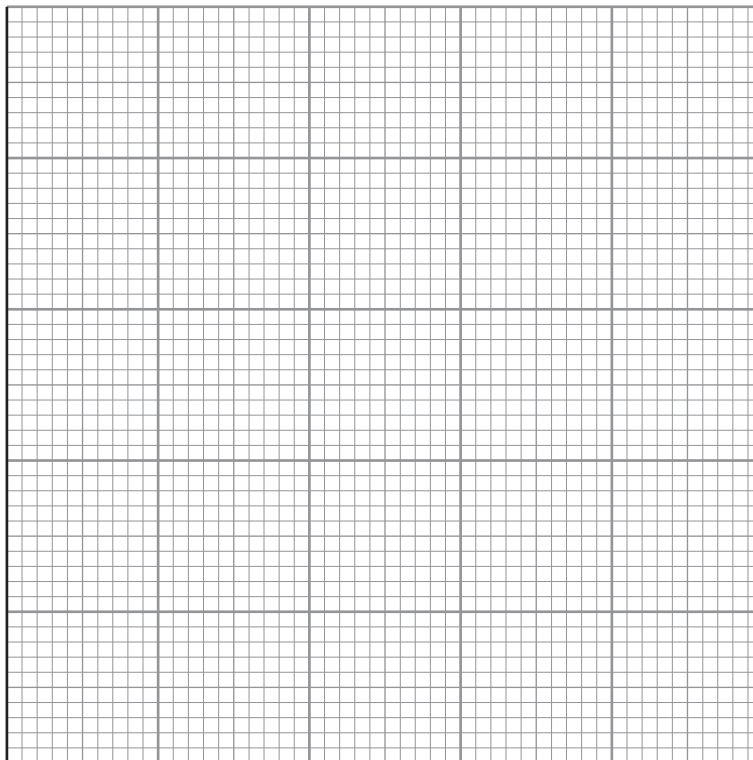
An experiment was carried out to determine the effect on the rate of reaction of varying the concentration of iodide ions. All other volumes and concentrations were kept constant. The results are shown in the table.

Concentration I^- / mol dm^{-3}	Time for appearance of blue-black colour / s	Rate / $\text{s}^{-1} \times 1000$
0.1	56	18
0.2	20	
0.3	18	
0.4	12	
0.5	10	

- (i) Use $\text{rate} = \frac{1000}{\text{time}}$ to calculate the rate for each experiment and complete the table. [1]



- (ii) On the axes below, plot the concentration of I^- against rate and draw a suitable line. [3]



- (iii) From the graph deduce the relationship between concentration of I^- and rate of reaction. [1]

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- (iv) **Use the graph** to calculate the time the reaction would take to turn blue-black using a 0.15 mol dm^{-3} solution of I^- . Show clearly how you obtained your answer. [2]

Time = s



- (v) For each experiment the rate was calculated using the time taken to produce excess iodine. Explain why this is only an approximation for the rate **as the reaction proceeds**. [2]

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- (b) (i) Draw a Boltzmann energy distribution curve. Label the axes. [2]

- (ii) Use this energy distribution curve to explain how catalysts affect the rate of a reaction. [2]

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